

Protect Yourself With This NOISE LEVEL MONITOR

NAVPREET SINGH TUNG

Presented here is an Arduino-based noise/sound level monitor that displays sound levels in decibels (dB) and corresponding voltage levels on an LCD. A noise monitoring system is used in many applications such as healthcare, defence, mining, robotics and so on.

This circuit is especially designed for noise monitoring in public places such as parties and marriage functions. Hearing capability of a person is around 80dB—sounds above this can cause hearing loss if high sound levels or volume are played for long.

Circuit and Working

The circuit diagram of the noise monitoring system is shown in Fig. 1. It is built around Arduino Uno board, 9V battery, 16x2 alphanumeric LCD (JHD162A), sound sensor module and a few other components.

When sound is detected by the microphone in the sound sensor module, corresponding voltage and sound levels are displayed in dB on LCD1. Sound sensor output is analogue voltage, which gets converted to digital through the inbuilt analogue-to-digital converter of the Arduino. Analogue voltage ranges from 0V through 5V, and the corresponding equivalent digital value 0 through 1023 is displayed on the serial monitor of Arduino IDE.

Sound sensor module. This device receives sound waves and converts them into electrical signals. It detects the sound intensity in ambient atmosphere.

The sound sensor module used in this project is shown in Fig. 2. It has a capacitive microphone that is sensitive to sound. Sound waves change capacitance and the corresponding voltage changes. Since change is quite weak, it needs to be amplified. LM393 is used as a power amplifier in the module. It has a built-in preset to adjust sensitivity of the analogue output pin (OUT).

Connect OUT pin (or S pin in some cases) of the sound sensor module to

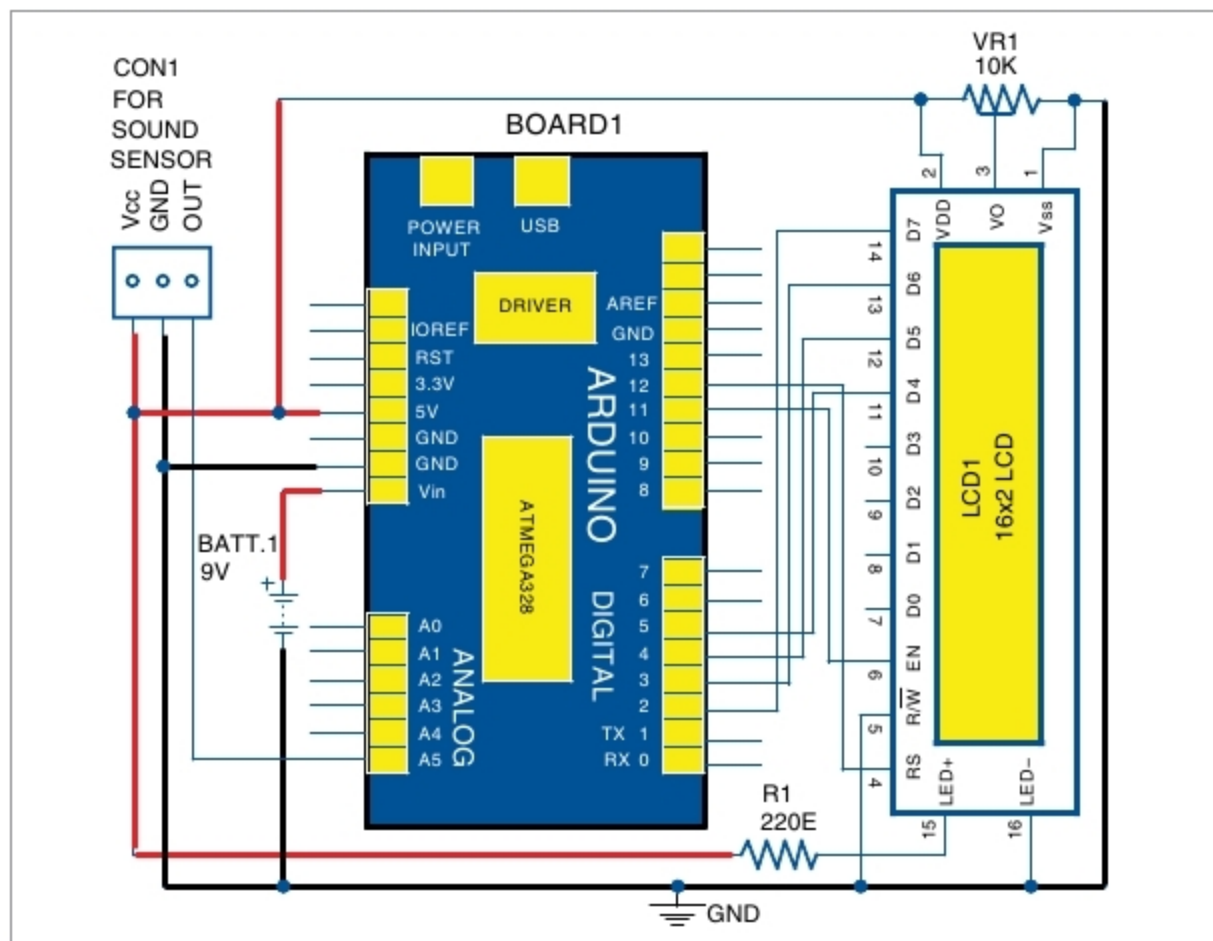


Fig. 1: Circuit diagram of the noise level monitor

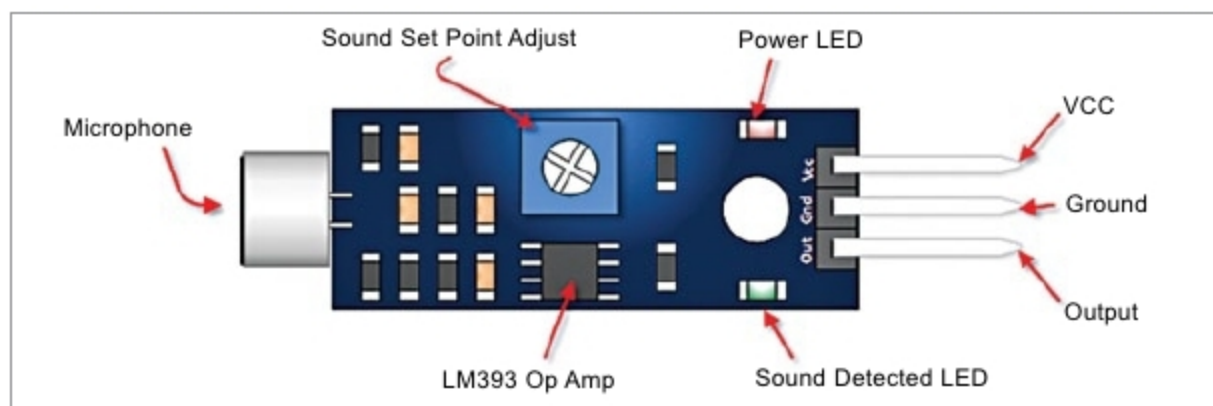


Fig. 2: Sound sensor module

analogue input pin of Arduino Uno. You will see the value of sound intensity on the serial monitor as well as on LCD1.

LCD module. 16x2 LCD module (LCD1) is a 16-pin LCD, which is easily available in the market. Refer to its datasheet to get pin details. A preset (VR1) is used to adjust the contrast of the LCD display.

Arduino Uno. Arduino Uno R3 is based on ATmega328 microcontroller (MCU). It contains fourteen digital input/output pins, six analogue input

pins, a 16MHz crystal oscillator, USB connection, power jack, ICSP header and reset button.

Programming is done using Arduino IDE. Analogue input pins A0 to A5 are available on Arduino Uno (Board1). When the software program (Audio_meter.ino) detects analogue signals on these input pins, corresponding signal is shown on the serial monitor of the PC and on LCD1.

Construction and testing

An actual-size PCB layout of the noise